# SW Engineering CSC648-848 Fall 2025

## Project Tutorall, Milestone 2 Part 1

Team 4
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| History | |
|---|---|
| Date Submitted | 11/01/2025 |
| Date Revised | |

## 1. Executive Summary

TutorAll is a web platform designed exclusively for San Francisco State University students to connect with qualified peer tutors across a variety of subjects. The goal is to simplify the tutoring process on campus by creating a centralized, secure, and easy-to-use website where students can browse and become verified SFSU tutors, view their subjects, hourly rates, and experience, then send a quick one-way message to request a session.

Many students struggle to find reliable academic help outside of class or don't know where to look for affordable tutoring options. Existing third-party services like Wyzant or Chegg Tutors are costly, cluttered, and don't focus on SFSU's unique academic environment. TutorAll solves this by creating a local, student-driven ecosystem that highlights SFSU tutors who already understand campus courses, professors, and grading styles.

The site's simplicity is its biggest strength. Users can search tutors by subject or SFSU class, see their listed rate, and contact them directly through a clean one-way messaging system that prevents spam and protects privacy. All payment details and scheduling are handled outside the platform, keeping TutorAll lightweight and focused purely on connection. The application is custom-tailored for SFSU and built with an intuitive interface aligned with the university's branding and accessibility standards.

Our development team is a group of senior computer science students passionate about improving the academic experience at SFSU. We are combining modern web technologies with user-centered design and feedback from students and faculty to deliver a platform that bridges the gap between those who need help and those who can provide it. TutorAll aims to make academic support at SFSU faster, safer, and more personal, a true student-for-students project.

2. **<u>List of main data items and entities</u>**

**Adverts**

- Location - string
- Price Range - a pair of double
- Verified - boolean
- Courses - list of courses
- Files - list of files - Optional
- User - integer

**Categories**

- Name - string
- Courses - list of courses

**Contributors**

- Name - string
- Image - image file
- Bio - string
- Links - list of key value pairs
- Role - string

**Courses**

- Name - string
- Professors - list of professors

**Files**

- Name - string
- File Type - string
- Upload Date - Date Time
- Uploader - integer

**Professor**

- Professor Courses - list of courses
- Name - string

**Users**

- Name - string
- Email - string
- Password - string
- Bio - string - Optional
- Education - list of strings - Optional
- Location - string - Optional
- Profile Picture - image file - Optional
- Links - list of key value pairs - Optional
- Active - boolean

**User Messages** (Updated from Messages)

- Contents - string
- Timestamp - Date Time
- Read - boolean
- Receiver_id - integer
- Sender_id - integer

3. **<u>Functional Requirements - prioritized</u>**

**- Priority 1**

**Unregistered User:**

1. Users shall be able to refine their search by class, professor, category, and price listing.
2. Users shall be able to view a list of tutors and detailed advert prior to registration.
3. Users shall be able to register for services.

**Registered Users:**

4. Users <u>shall</u> be able to message other registered users directly.
5. Users <u>shall</u> be able to post tutor forms for approval.
6. Users <u>shall</u> be able to edit, remove, and view listings after approval.
7. Users <u>shall</u> be able to see pending tutor listings.
8. Users <u>shall</u> be able to provide documentation for tutor listings.
9. Users <u>shall</u> be able to deactivate their account with confirmation.
10. Users <u>shall</u> be able to add multiple classes to one tutor listing.
11. Users <u>shall</u> be able to dictate a meeting address, visible through Google Maps API.
12. Users <u>shall</u> be able to cancel unapproved listings
13. Users <u>shall</u> be able to view an organized inbox of internal messages.

**Admin:**

14. Admin Users <u>shall</u> be able to approve tutor listings
15. Admin Users <u>shall</u> be able to delete offending content
16. Admin Users <u>shall</u> be able to deactivate users

**- Priority 2**

**Unregistered Users:**

17. Users <u>shall</u> be able to view a tutor's personal profile.

**Registered Users:**

18. Users <u>shall</u> be able to provide personal information via user profile.

**- Priority 3**

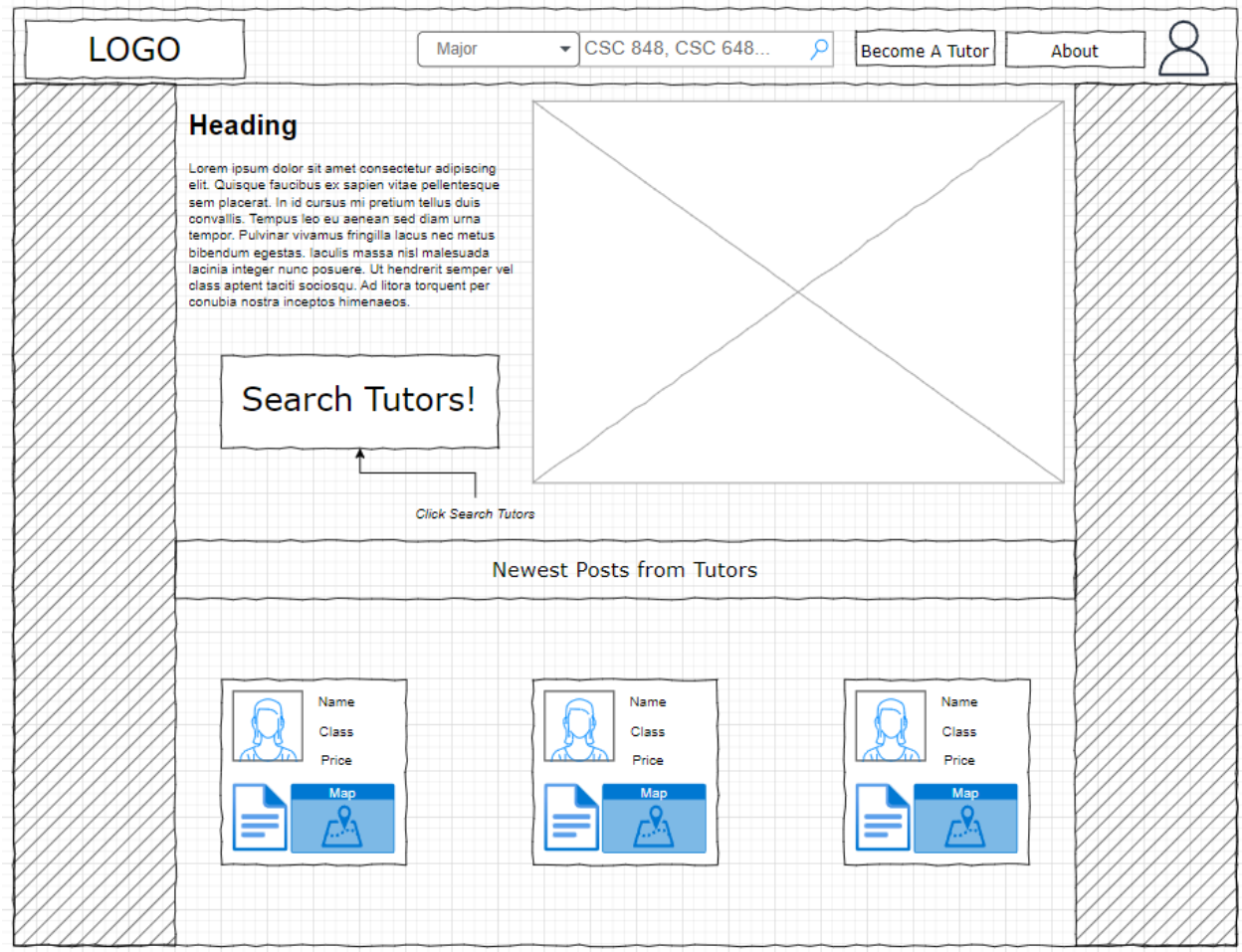
**Registered Users:**

19. Users shall be able to provide sanitized links in their biography.

4. **<u>UI Storyboards for each main use case</u>**

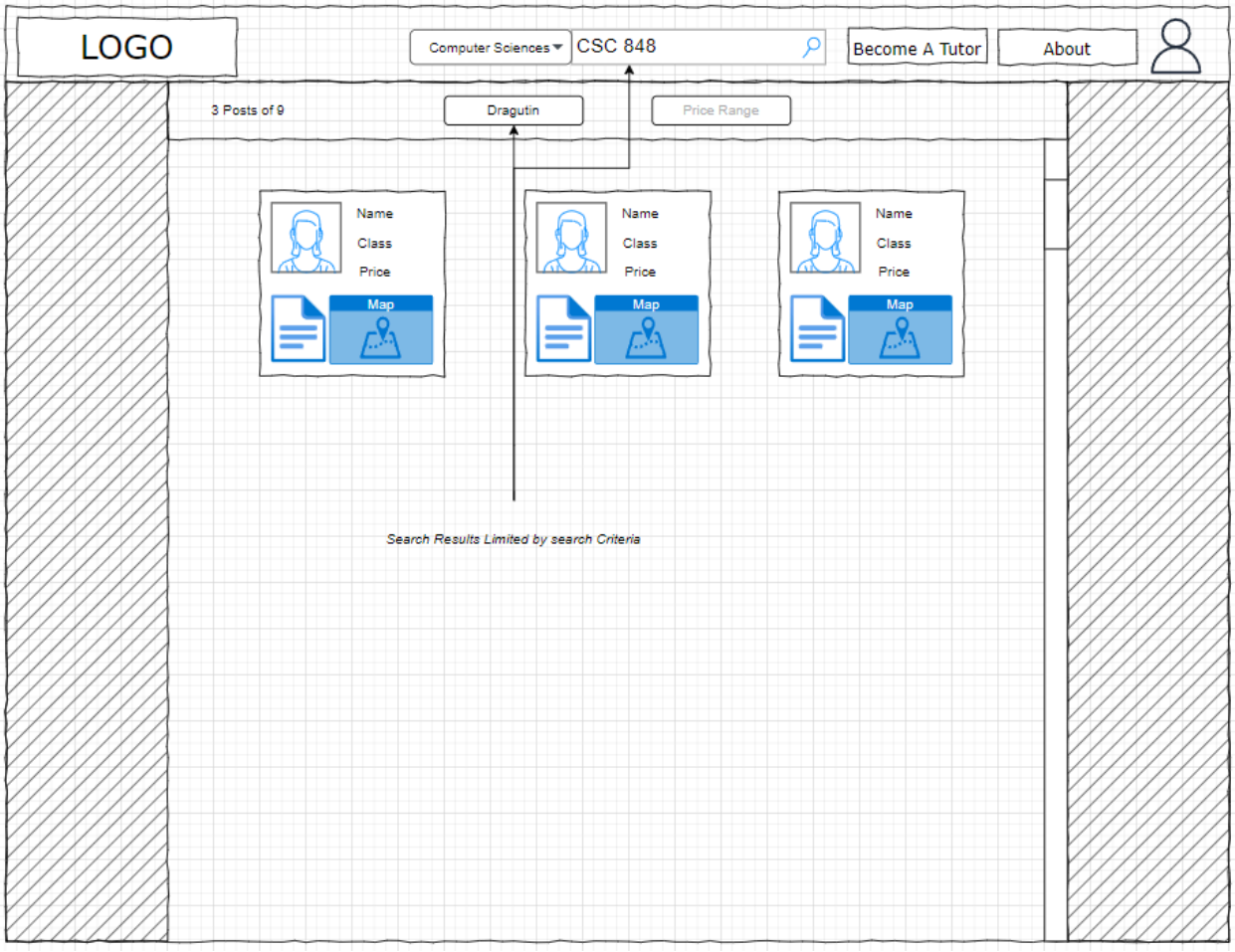
**<u>Use Case 1:</u>**

I, as a student, want to navigate to the main webpage to find tutors quickly. I want to find a tutor for a specific class from a specific professor, and I want to know if it's within my budget. Last of all, I do not want to register to see if tutors are available



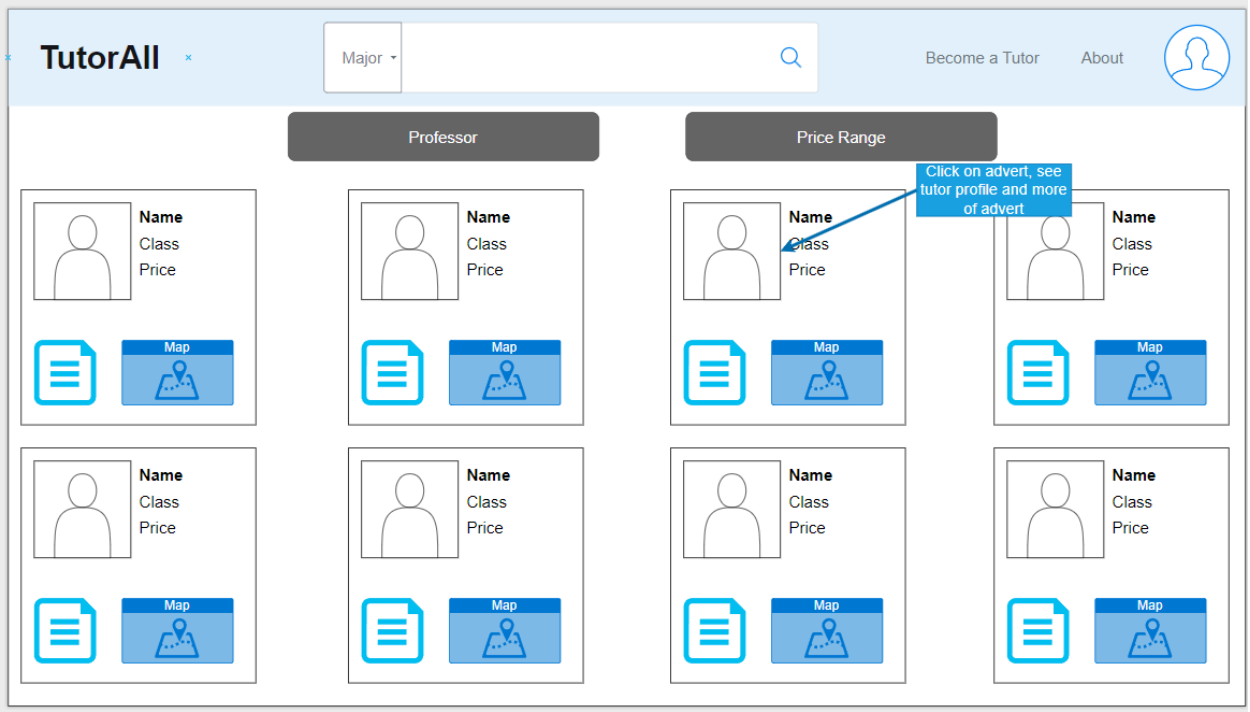
LOGO
Major
CSC 848, CSC 648...
Become A Tutor
About
Professor
Price Range
Name
Class
Price
Map

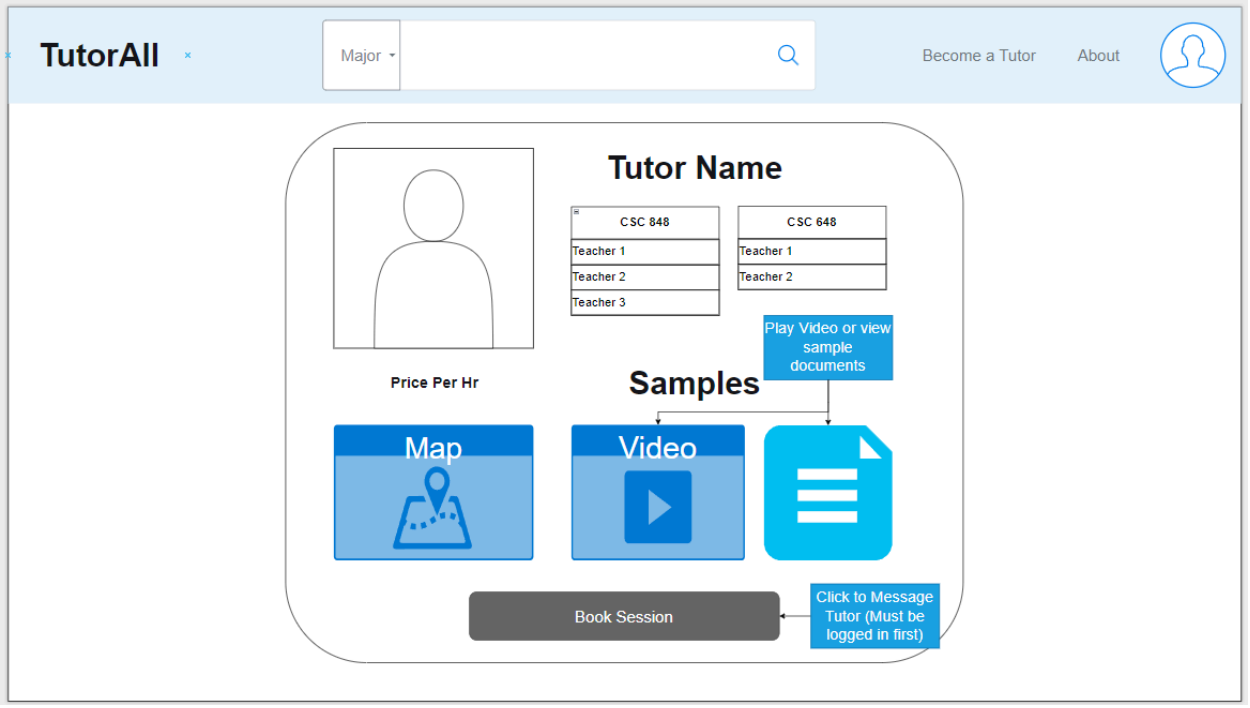
LOGO
Major
CSC 848, CSC 648...
Become A Tutor
About
Professor
Price Range
Refine Search by filling in text.
Name
Class
Price
Map


LOGO
Computer Sciences
CSC 848
Become A Tutor
About
3 Posts of 9
Dragutin
Price Range
Name
Class
Price
Map
Name
Class
Price
Map
Name
Class
Price
Map
Search Results Limited by search Criteria

**Use Case 2:**

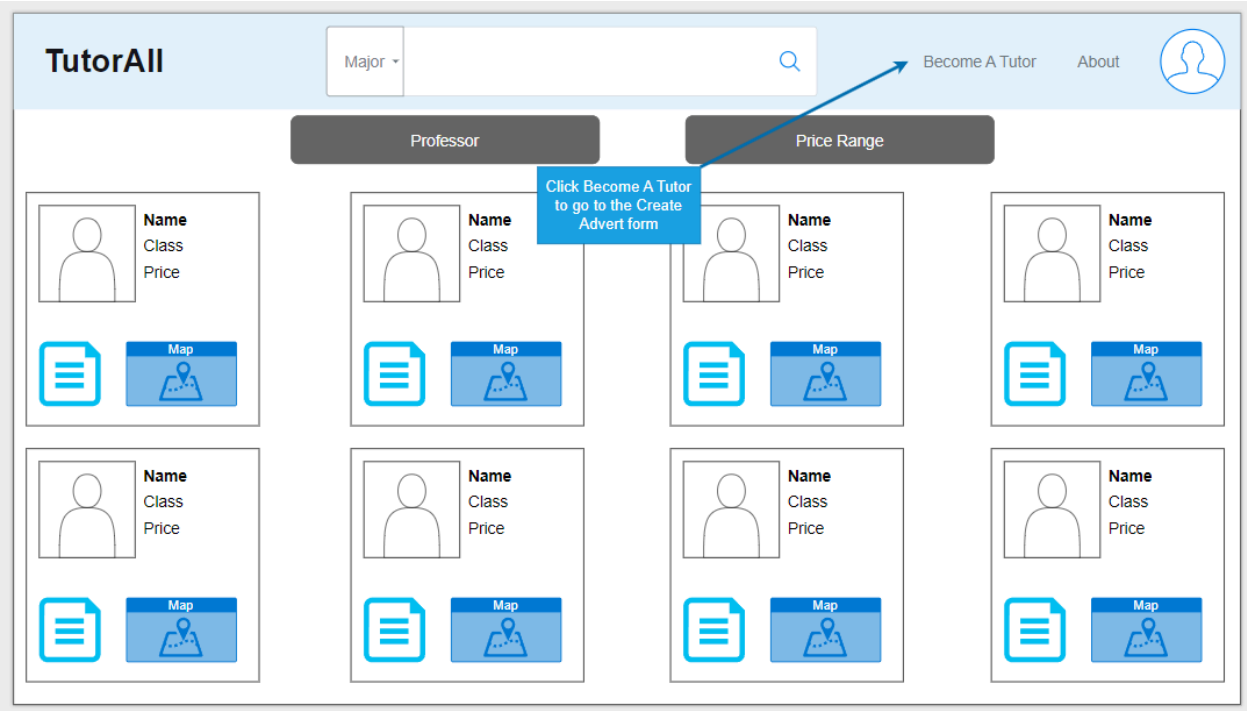
I, as a student, want to be able to know more about the tutor before sending them a message, to make sure that I am making the right decision to message them.

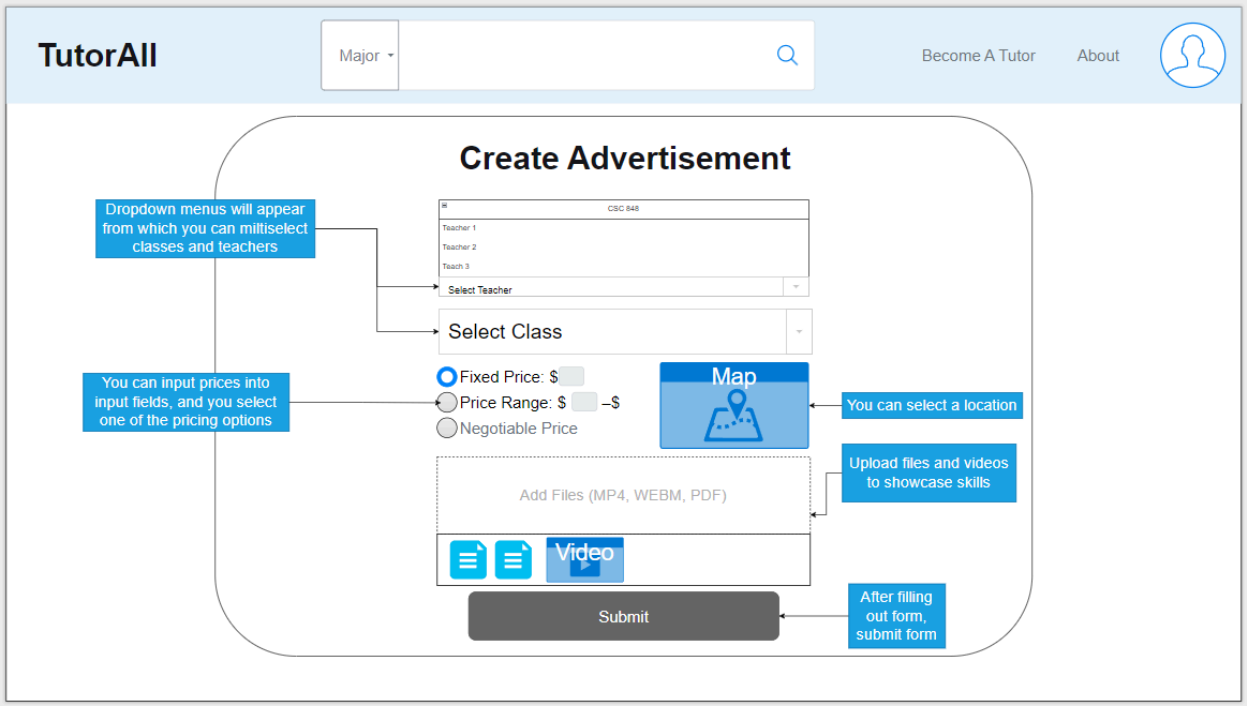


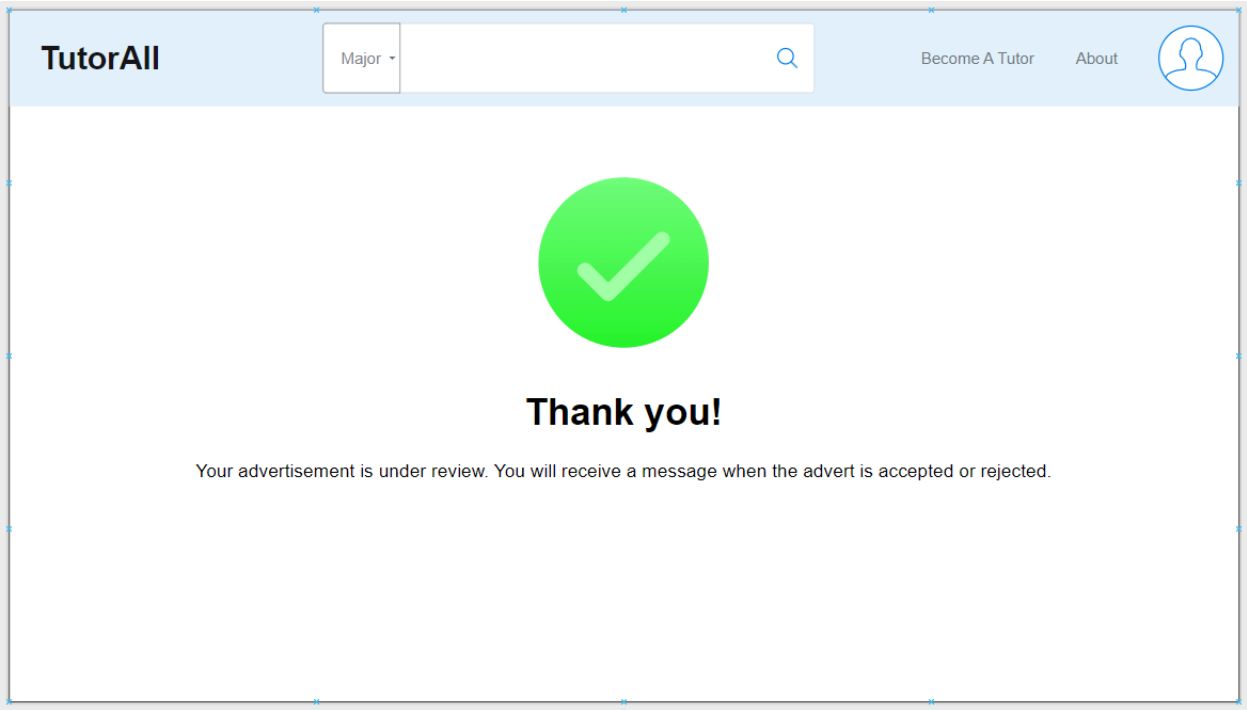



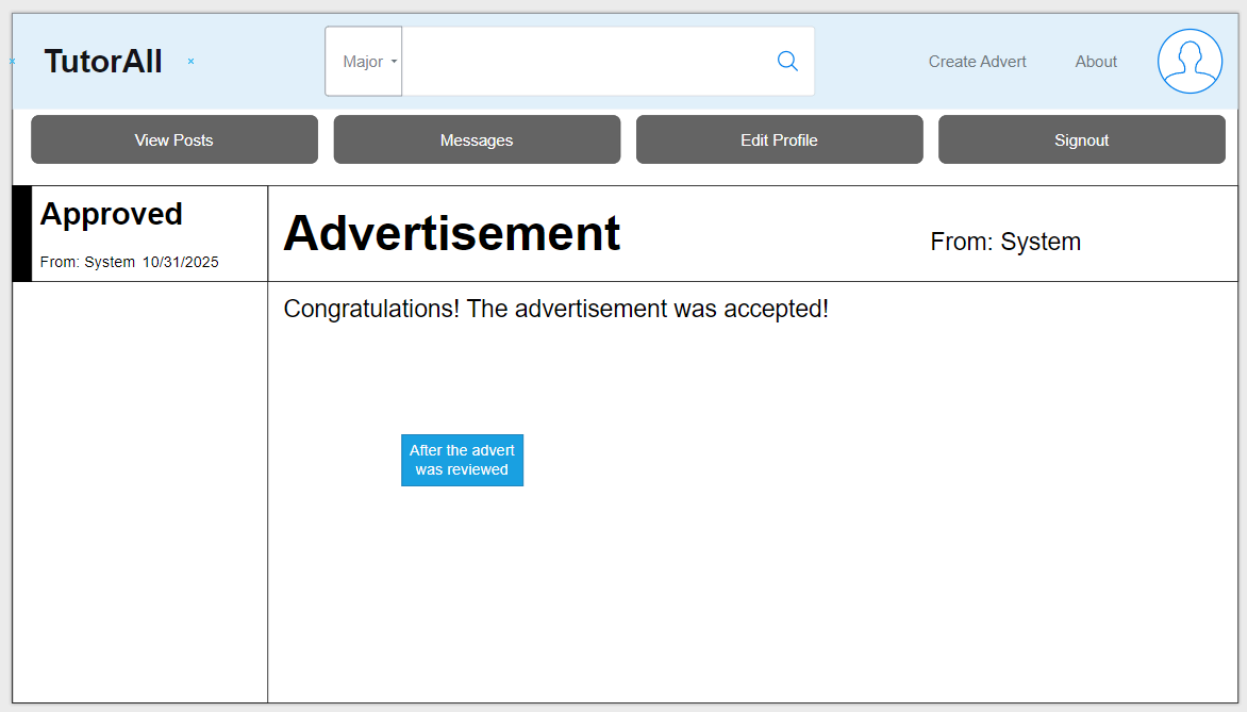
**Use Case 3:**

I, as a tutor, want to be able to submit a form for my tutor services for authorization. I want to be able to provide a class and teacher, or multiple classes and associated teachers. I want to be able to determine my price, price range, or be willing to negotiate. I want to be able to provide my location, or a location for meeting. I wish to be able to upload a file (pdf or video) to showcase my skills. I do not want to have to provide my name or picture for my listing. When I am done, I wish to submit my form for review, and be notified by message when it is verified (or rejected).



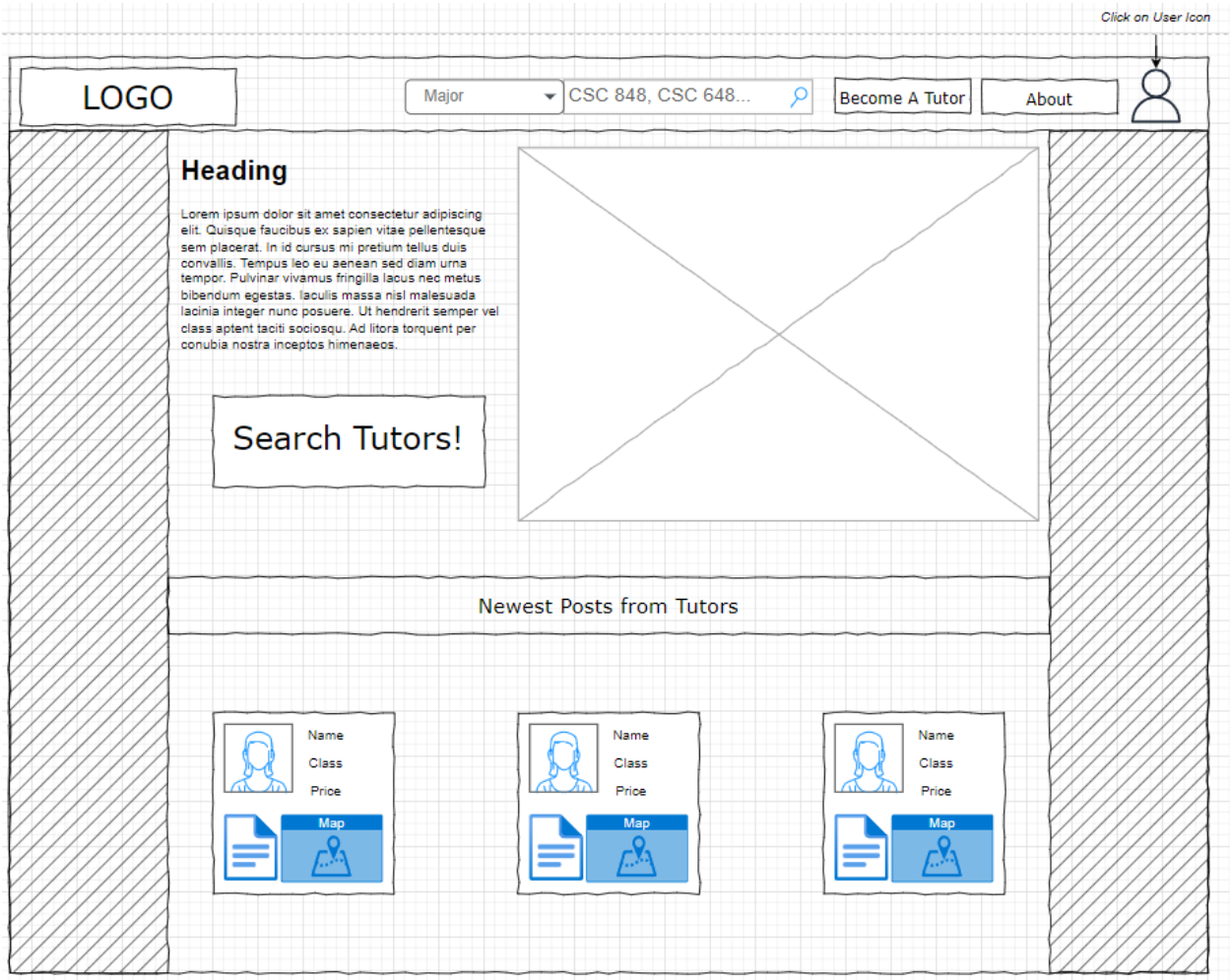

TutorAll
Major
Become A Tutor
About
Create Advertisement
Dropdown menus will appear from which you can miltiselect classes and teachers
CSC 848
Teacher 1
Teacher 2
Teach 3
Select Teacher
Select Class
You can input prices into input fields, and you select one of the pricing options
Fixed Price: $
Price Range: $ –$
Negotiable Price
Map
You can select a location
Upload files and videos to showcase skills
Add Files (MP4, WEBM, PDF)
Video
Submit
After filling out form, submit form

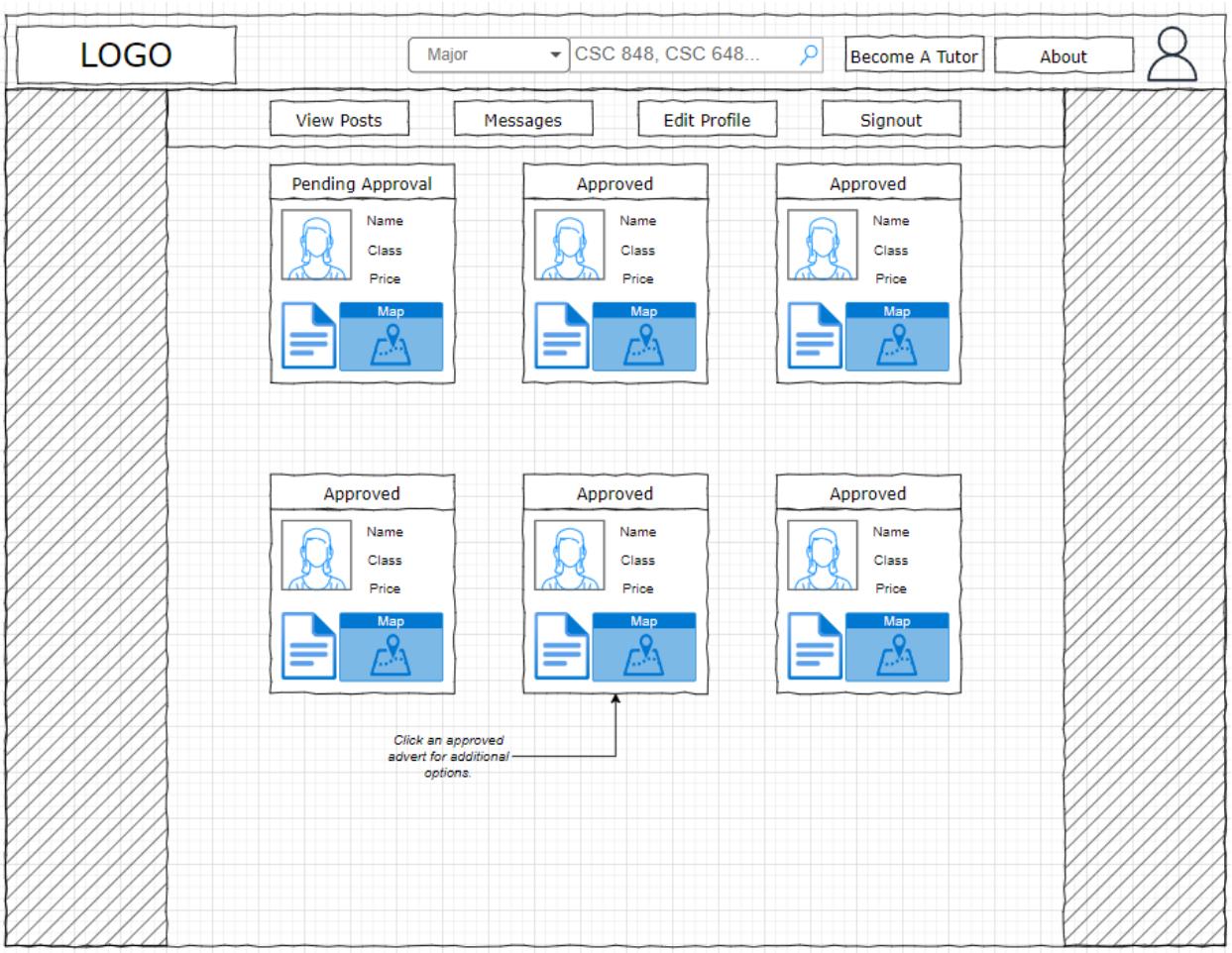

TutorAll
Major
Become A Tutor
About
Thank you!
Your advertisement is under review. You will receive a message when the advert is accepted or rejected.


TutorAll
Major
Create Advert
About
View Posts
Messages
Edit Profile
Signout
Approved
From: System 10/31/2025
Advertisement
From: System
Congratulations! The advertisement was accepted!
After the advert was reviewed

**Use Case 4:**

I, as a tutor, wish to be able to track, edit, and remove my listings, and check messages from a convenient screen.




LOGO
Major
CSC 848, CSC 648...
Become A Tutor
About
View Posts
Messages
Edit Profile
Signout
Pending Approval
Approved
Approved
Approved
Approved
Approved
Name
Class
Price
Map
Click an approved advert for additional options.

LOGO
Major
CSC 848, CSC 648...
Become A Tutor
About
View Posts
Messages
Edit Profile
Signout
Pending Approval
Approved
Approved
Approved
Approved
Approved
Name
Class
Price
Map
View
Edit
Remove
Click again to hide options.

LOGO
Major
CSC 848, CSC 648...
Become A Tutor
About
View Posts
Messages
Edit Profile
Signout
Pending Approval
Name
Class
Price
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Class
Price
Map
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Name
Class
Price
Map
Click pending for more options
Approved
Name
Class
Price
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Name
Class
Price
Map

LOGO
Major
CSC 848, CSC 648...
Become A Tutor
About
View Posts
Messages
Edit Profile
Signout
Pending Approval
Name
Class
Price
Map
View
Cancel
Approved
Name
Class
Price
Map
Approved
Name
Class
Price
Map
Click again to close options
Approved
Name
Class
Price
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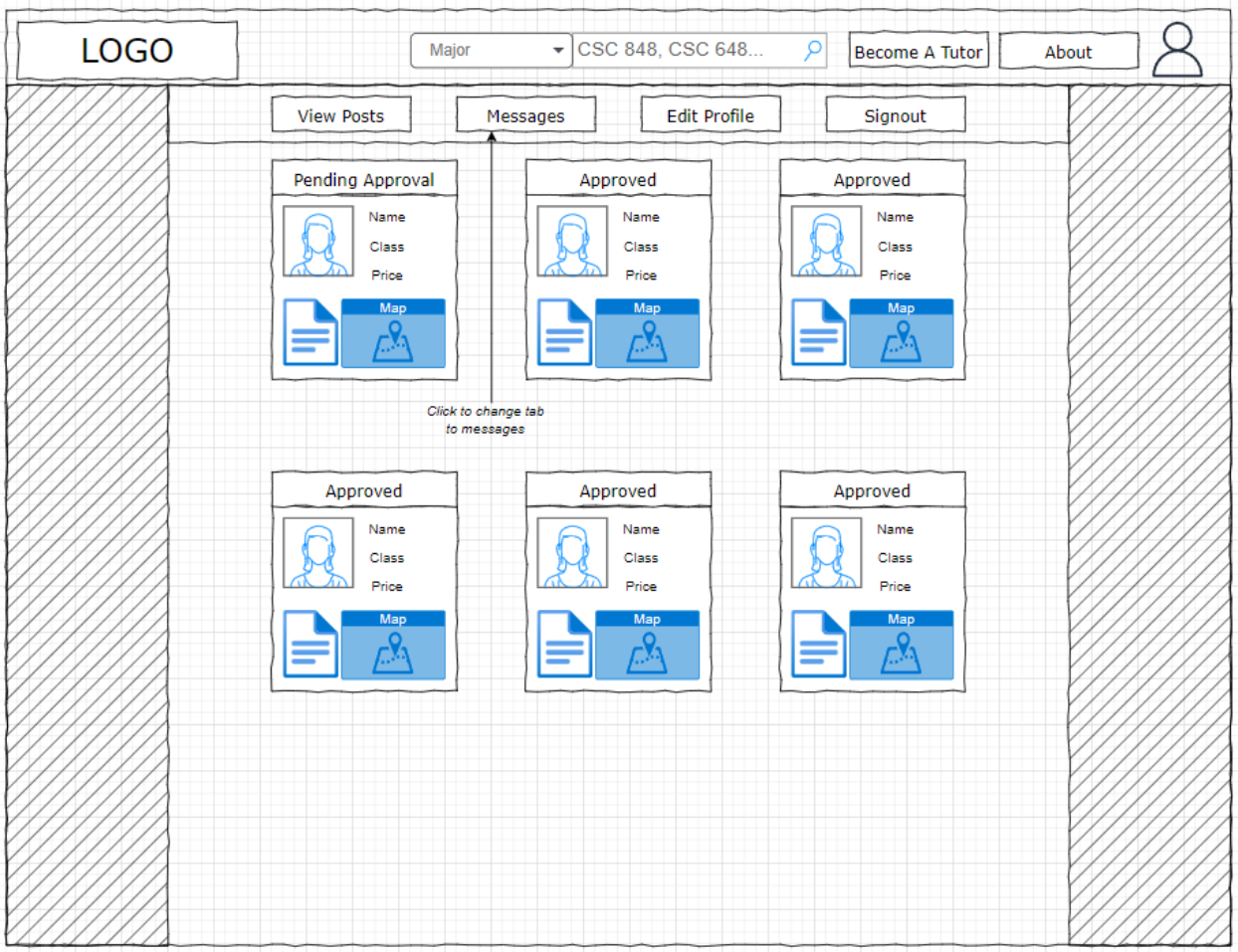
Name
Class
Price
Map


LOGO
Major
CSC 848, CSC 648...
Become A Tutor
About
View Posts
Messages
Edit Profile
Signout
Pending Approval
Name
Class
Price
Map
Approved
Name
Class
Price
Map
Approved
Name
Class
Price
Map
Click to change tab
to messages
Approved
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Name
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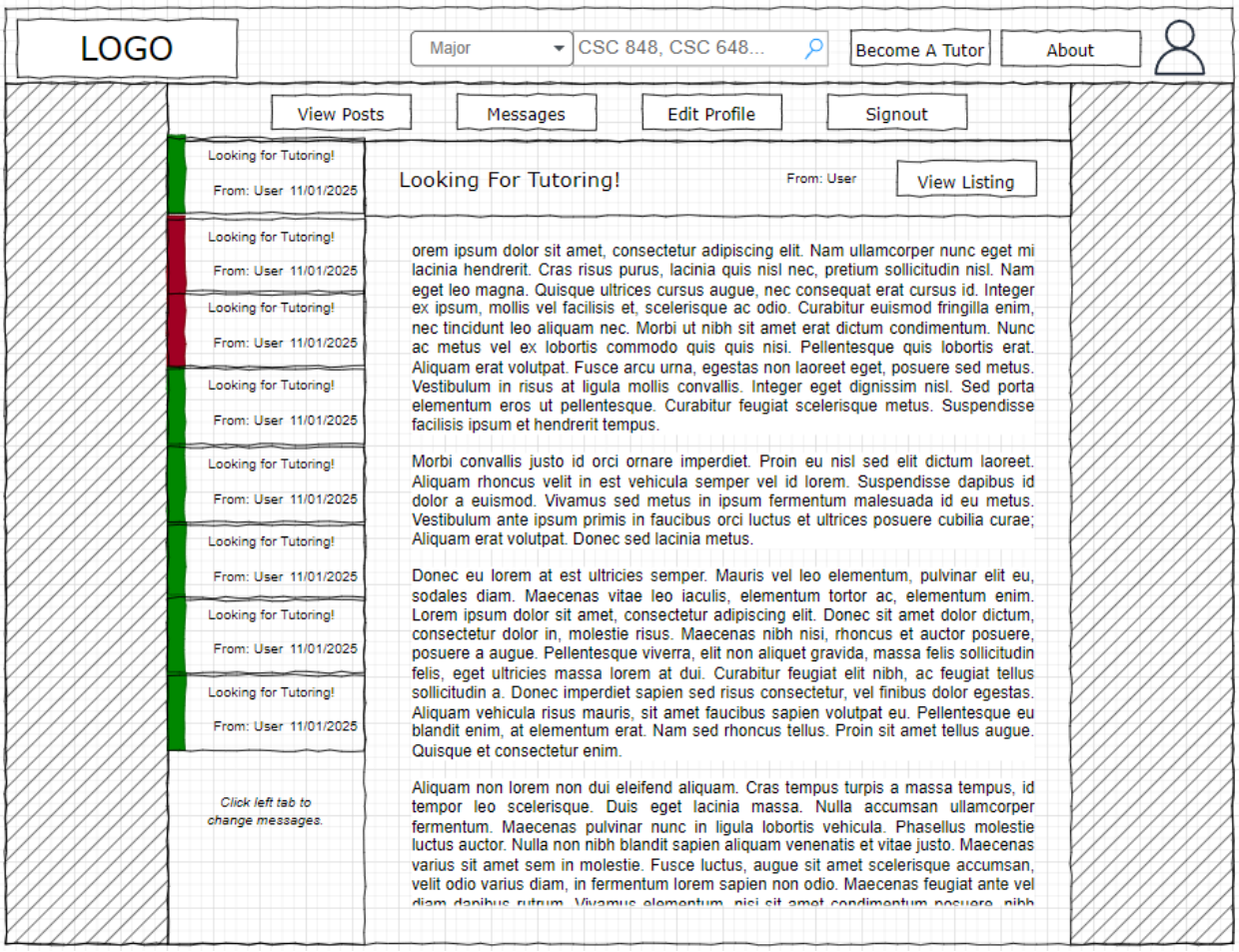
Map

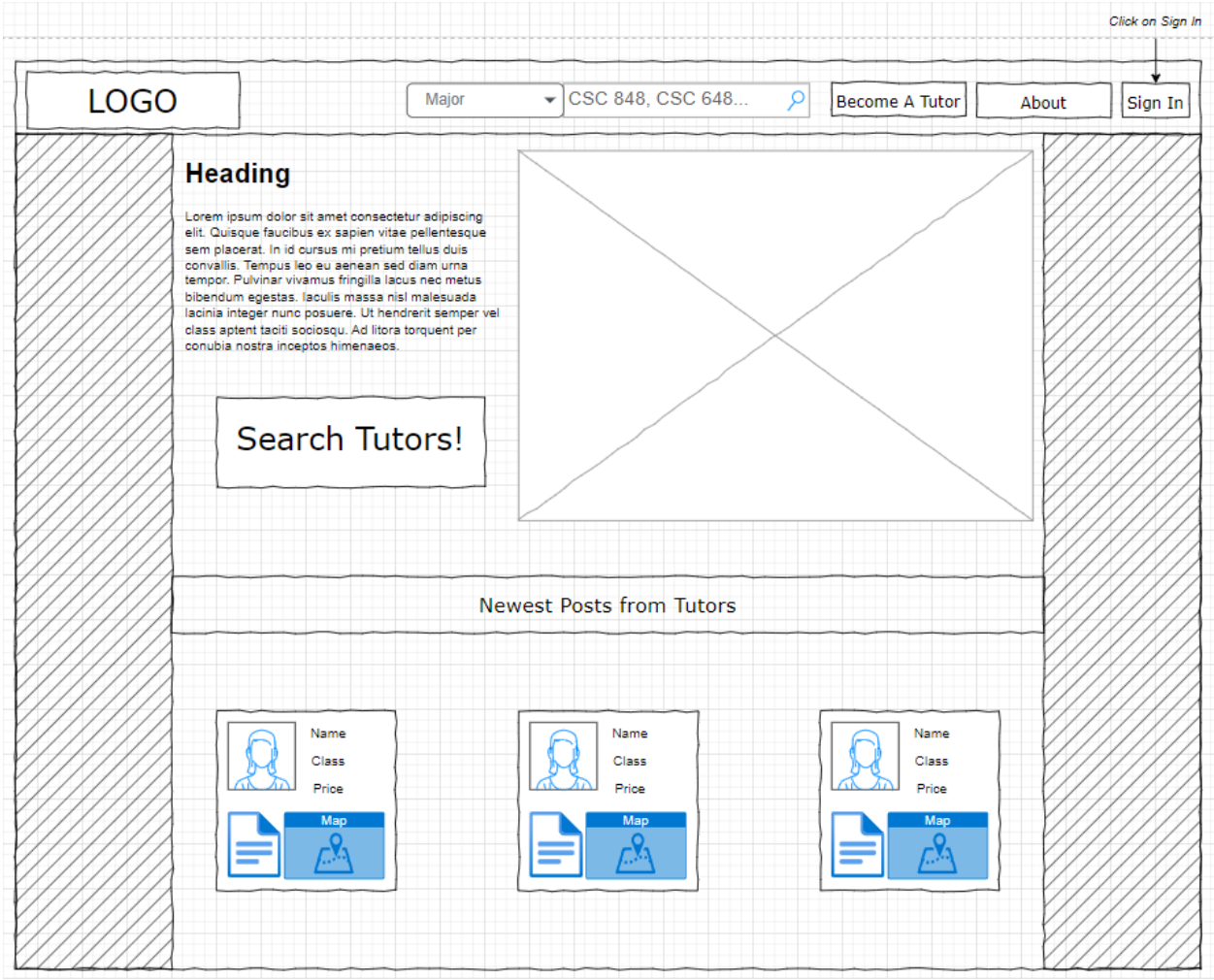


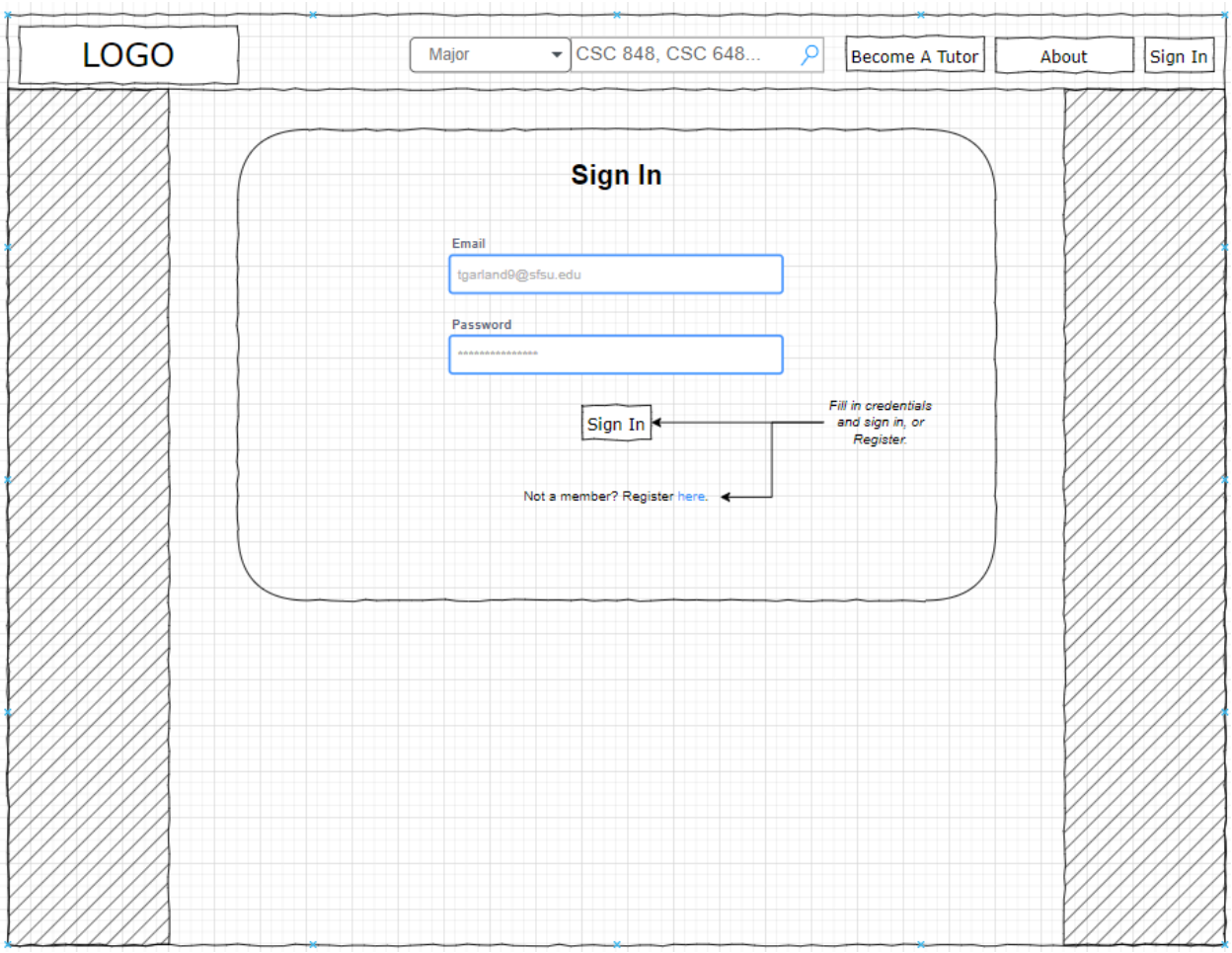

**<u>Use Case 5:</u>**

I, as an Admin, want to be able to approve and remove content, and disable offending accounts, prior to them going live.

- We will be using DRF for admins to interact with the DB similar to Workbench. Here, an admin can login using their credentials, and directly approve and remove content.

## Use Case Login / Registration:




LOGO
Major
CSC 848, CSC 648...
Become A Tutor
About
Sign In
Sign In
Email
tgarland9@sfsu.edu
Password
Sign In
Fill in credentials and sign in, or Register.
Not a member? Register here.

## 5. High level Architecture, Database Organization

*DB Organization:*

**Adverts**- Adverts is the tutor listings that are displayed on the website to see if it matches the users needs

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| id | int | NO | PRI | NULL | |
| location | varchar(255) | NO | | NULL | |
| price_range | decimal(65,2) | NO | | NULL | |
| verified | tinyint(1) | NO | | NULL | |
| courses_id | int | NO | MUL | NULL | |
| files_id | int | NO | MUL | NULL | |
| users_id | int | NO | MUL | NULL | |

**Categories**- The type of courses that it is under

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| id | int | NO | PRI | NULL | |
| name | varchar(255) | NO | | NULL | |
| courses_id | int | NO | MUL | NULL | |

**Contributors**- The information of the team that created the website

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| id | int | NO | PRI | NULL | |
| name | varchar(255) | NO | | NULL | |
| image | varchar(100) | NO | | NULL | |
| bio | longtext | NO | | NULL | |
| links | json | NO | | NULL | |
| role | varchar(255) | NO | | NULL | |

**Courses**- The type of classes that are on the website and needed

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| id | int | NO | PRI | NULL | |
| name | varchar(255) | NO | | NULL | |
| professor_id | int | NO | MUL | NULL | |

**Files**- The model where the adverts can show off their portfolio alongside with the adverts

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| id | int | NO | PRI | NULL | |
| name | varchar(255) | NO | | NULL | |
| file_type | varchar(100) | NO | | NULL | |
| upload_date | datetime(6) | NO | | NULL | |
| uploader_id | int | NO | MUL | NULL | |

**Professor**- Shows all of the teacher that is teaching a particular course

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| id | int | NO | PRI | NULL | |
| professor_courses_id | int | NO | MUL | NULL | |
| name | varchar(255) | YES | | NULL | |

**Users**- The users are the ones that registered in the website with a quick description about them

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| id | int | NO | PRI | NULL | |
| name | varchar(255) | NO | | NULL | |
| email | varchar(255) | NO | | NULL | |
| password | varchar(255) | NO | | NULL | |
| bio | longtext | NO | | NULL | |
| education | varchar(255) | NO | | NULL | |
| location | varchar(255) | NO | | NULL | |
| profile_picture | varchar(100) | NO | | NULL | |
| links | json | NO | | NULL | |
| active | tinyint(1) | NO | | NULL | |

**User_messages**- Registered users would send messages to ask for help on a particular subject

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| id | int | NO | PRI | NULL | |
| contents | longtext | NO | | NULL | |
| timestamp | datetime(6) | NO | | NULL | |
| read | tinyint(1) | NO | | NULL | |
| receiver_id | int | NO | MUL | NULL | |
| sender_id | int | NO | MUL | NULL | |

*Media Storage*

We will be storing the media files as a file on the server. Storing files as blobs would make the database take up too much storage if a person decided to upload a large file in the database. By storing it as a file, it would leave up to AWS or nginx to properly display the files. We will also put file size restrictions to prevent the files from taking too much storage in the server.

*Search/filter architecture and implementation*

Our tech stack includes Django, it already has a built-in Object Relational Mapping (ORM) that does most of the queries before updating the database. Filtering and searching with the Django ORM will call the .filter(field_name_iexact=<string>),iexact is looking for a string that is case insensitive which is similar to using the %<string> or %<string>% in MySQL. The data items that will be searchable are the courses, category, professor, and the adverts.

6. **Identify actual key risks for your project at this time**

Skill Risk - Google Maps
No members of the team have integrated Maps API into prior projects. This could prove to be a risk to implement and utilize it. However, we intend to solve this through dedicating time to review tutorials and API documentation. Additionally, we will be using extra time during our meetings to further discuss how we desire to use Google Maps API as a part of our project to ensure we're aligned on its use.

7. **Project management**

For Milestone 2, our team implemented a structured and collaborative approach to project management to ensure the timely and organized completion of all assigned tasks. We utilized GitHub Issues as our primary tool for task allocation, tracking, and progress monitoring. Once our team lead reviewed the milestone document, they divided the deliverables into specific components and created corresponding issues for each section. These issues were then assigned to individual team members along with clear descriptions and deadlines. This allowed every member to have a defined role and provided the entire team with visibility into the progress and status of each task through GitHub's integrated dashboard.

To maintain strong coordination and accountability, our team meets twice weekly, once after class on Wednesdays and again on Fridays. During these meetings, we review the progress made throughout the week, address any obstacles, and plan the next set of objectives. These consistent check-ins serve as both progress reviews and mid-point checkpoints before each submission, ensuring adherence to our established timeline. In addition to scheduled meetings, our group communicates actively throughout the week using GitHub comments and group messaging to clarify requirements, share updates, and provide feedback in real time.

As we move forward with future milestones, we plan to strengthen this management framework by maintaining clear task ownership and consistent communication between the front-end and back-end teams. Each subteam will continue to operate with a degree of independence while collaborating on shared interfaces to maintain system consistency. This structured approach promotes accountability, efficiency, and effective collaboration, ensuring that all deliverables are completed with

high quality and within the designated timeframe.

8. **<u>Use of genAI tools like ChatGPT and copilot</u>**

**Matthew A. Davis:**
Did not use Gen AI.

**Tan Huynh:**
Did not use Gen AI.

**Anmol Tadikonda:**
Did not use Gen AI.

**Justine Tenorio:**
Did not use Gen AI.

**Zainab Abbasi:**
• UI Storyboarding • HIGH
• Used it to figure out panels in storyboards
• Asked ChatGPT what the panels should be

9. **<u>Team Lead Checklist</u>**

So far all team members are fully engaged and attending team sessions when required - **OK**

Team ready and able to use the chosen back and front end frameworks and those who need to learn are working on learning and practicing - **OK**

Team reviewed suggested resources before drafting Milestone 2 - **On Track**

Team lead checked Milestone 2 document for quality, completeness, formatting and compliance with instructions before the submission - **On Track**

Team lead ensured that all team members read the final Milestone 2 document and agree/understand it before submission - **OK**

Team shared and discussed experience with genAI tools among themselves - **OK**